1. Clinical Question & Core Hypothesis

Clinical Question

Why do some tumors with favorable immune architecture still fail to achieve better clinical outcomes?

Working Hypothesis

The prognostic impact of high Gleason Grade Group 4–5 prostate cancers, characterized by predominant cribriform or poorly formed gland architecture, is mechanistically mediated by the tumor microenvironment. Specifically, a high stromal-to-tumor ratio (TSR) creates an ECM-rich, desmoplastic niche that enhances local invasion and accelerates progression after local therapy. Consequently, among Grade Group 4–5 tumors, those with high TSR will exhibit a significantly shorter progression-free interval (PFI) after local therapy compared with those with low TSR. In tumors with the same high-grade architecture but low TSR, the rate of progression will be comparatively attenuated, indicating that TSR modulates the risk conferred by cribriform morphology. Anchor genes are not required for this primary hypothesis but may be incorporated as secondary modifiers if available. The hypothesis is testable by stratifying patients by Grade Group and TSR, then evaluating the interaction effect on PFI with appropriate adjustment for clinical and pathologic confounders.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Primary endpoint	PFI

2. Why This Gene — Not Others

3. Clinically Testable Predictions

Prediction 1: associated with worse outcomes. However, substantial heterogeneity exists within the same Grade Group, suggesting additional biological determinants.

Prediction 2: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions